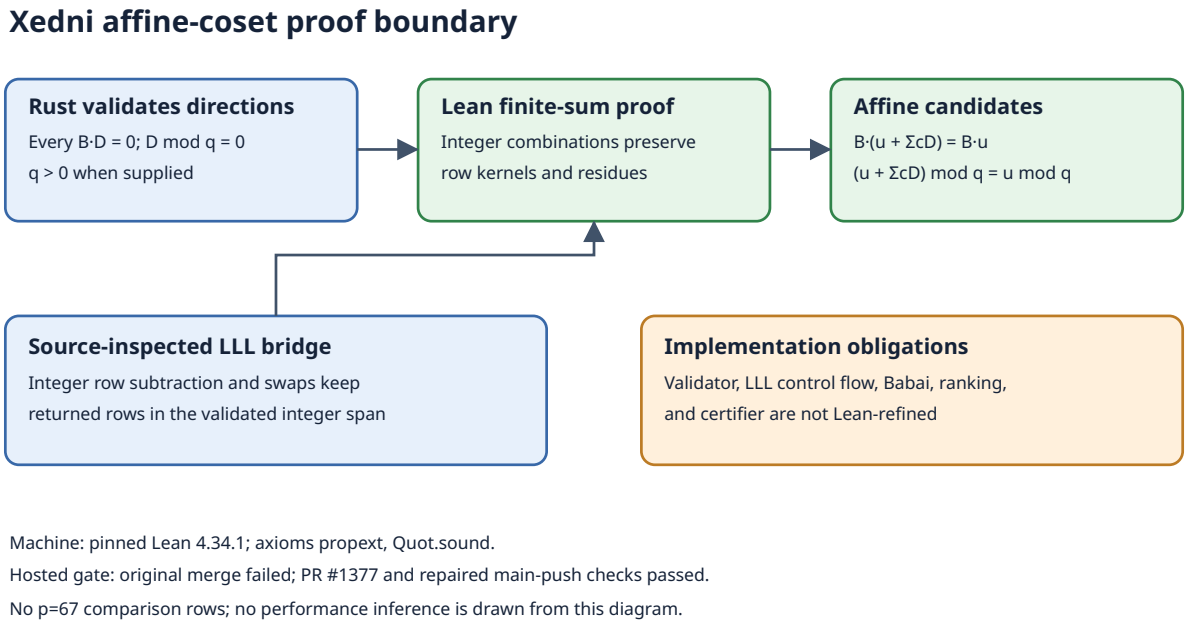
Xedni affine-coset Lean review

Technical packet, 2 November 2026

Scoped result. For finite integer row and direction families, if each direction annihilates every incidence row and is coordinatewise zero modulo positive q, every integer recombination preserves those premises. Each affine candidate $u + \sum_i c_i D_i$ has the same row dot products and residues as u. Three named Lean declarations pass the pinned Lean 4.34.1 checker; their axiom lists contain only `propext` and `Quot.sound`.



Independent review. Blind algebra task `TASK-20261102-1a2b3c` re-derived the finite-sum invariant without producer source, including separate failed controls for the row-kernel and residue premises. Source task `TASK-20261102-4d5e6f` traced the Rust validation, candidate loop, integer LLL row operations, certifier, receipt, and workflow. Both filed separate attestations; the shared independence checker found zero declared violations after a documented cross-repository field-name adapter.

Workflow finding and repair. The original merged workflow at cryptanalysis `749d807` failed on first-use elan setup even though Lean returned zero. Repair PR #1377 added explicit toolchain download and bound `lattice.rs` in the source hash and workflow paths. All 28 applicable checks passed at head `937b387`, including hosted Lean run 41507496535. The repair merged as `15d5263`; its main-push Lean run passed. The earlier failure remains historical evidence.

Claim separation. The Lean algebra is machine checked and faithful to the conditional candidate invariant. The Rust LLL span property is source-inspected, not a formal refinement. Parser behavior, full validator and LLL control flow, Babai choice, ranking, and certifier remain distinct implementation obligations. The frozen $p = 67$ diagnostic has no run rows because checkpoint upload was denied, so no performance curve is inferred.

Source bindings. Producer commit `d243c3c`; Lean SHA-256 `c39148ea...e757`; Rust candidate SHA-256 `b08c9484...ac96`; LLL SHA-256 `0687b6c0...0b`. The full hashes, per-claim table, controls, and report links are in the adjacent `README.md`. The cross-field degree graph was checked against its CSV and left unchanged because this proof changes no graph input.